


Sensors AND Transducers

Generalized Measuring System

Input parameter under measurement

↓
Sensor / Transducer

↓
Signal Conditioning

↓
Data Processing

↓
Transmission for Telemetry

↓
Output Device

Minor Project

→ PT Cells

→ Microprocessor
based temperature
compensation

Q. Discuss with neat diagram the concept of generalised measuring system.

Transducers :- A device that is used to convert one form of energy to another form.

All sensors are transducers but all transducers are not sensors.

Q. Classify sensors based on i) Activity, ii) Position of measurement, iii) Signal, iv) Principle of operation

Transducer Classification :-

1) According to activity

i) Energy Converter (Active)

- a) magnetic induction
- b) Thermoelectric
- c) photoelectric
- d) Piezoelectric

ii) Energy Controller (Passive)

- a) Resistive
- b) Inductive
- c) Capacitive

2) According to Output Signal

i) Analog

ii) Digital

3) According to working principle

- i) Mechanical
- ii) Thermal
- iii) Magnetic
- iv) Radiation

4) According to position of measurement

- i) Primary
- ii) Secondary

Q. State different factors to be considered for selecting sensors in mechatronics application

Q. Conceptual $\rightarrow$ Block Diagram

Factors influencing choice of Transducers

* Size & Weight

Ex:- Aerial drones (Need to be light & small)

* Shape

Ex:- Underwater / Underground pipes.

* Sensitivity

Ex:- For accurate measurements

* Accuracy

* Frequency response

* Transient response (Step change or sudden change)
 T_d , T_r , T_s , Peak Overshoot

* Static characteristics

Resolution :- Clarity (Smallest change input detected)

Accuracy :- Error

Precision :- Repetability (Giving 10V again & again)

- * **Ruggedness**
Shock absorber
- * **Reliability** (MTBF, MTTR)
Mean Time Before Failure
Mean Time To Repair
Repairing time of failure
- * **Stability**
Oscillations
- * **Environmental Capability**
Military Grade (withstand huge variations of temperature, pressure, humidity etc.)

Measurement

Pressure Transducers :-

- * $P = F/A$
- * Inferentially measure variables like **Flow & level**.
(indirect)
- * Determines boiling point of liquids.
- * Fluids exert pressure on walls of vessels equally in all directions.
- * Following considerations during pressure measurement
 - i) Pressure influenced by position within a static fluid but at a given position is independent of direction.

ii) Pressure is unaffected by the shape of the confining boundaries.

* Unit :- N/m^2 (Pascal)
Psi (Pounds per square inches)
bar

* Vacuum pressure measurement (Below atmosphere)
Ionization gauges

* Gauge pressure measurement (Above atmosphere)

$$\text{Gauge Pressure} + \text{Atmospheric Pressure} = \text{Pressure due to fluid itself} = \text{Absolute fluid pressure}$$

$$\text{where } 1 \text{ atm} = 14.696 \text{ psi} = 101.325 \text{ KPa}$$

* Pressure measurement methods

i) Elastic pressure transducers

ii) Primary pressure transducers (Physical $\rightarrow$ Physical)
Ex: pressure $\rightarrow$ displacement

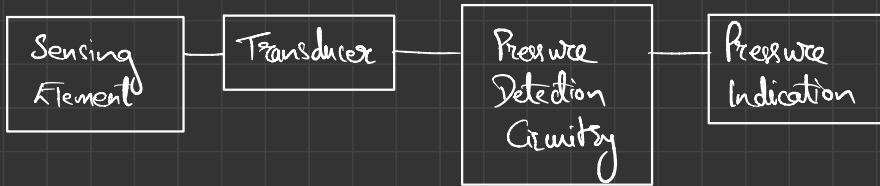
iii) Manometer method

iv) Pressure measurement by measuring vacuum

v) Elastic pressure transducers

vi) Pressure measurement by balancing forces produced on a known area by a measured force.

Electric Pressure Transducers



Analog signal transmission is in standard **current** range of 4mA to 20mA.

Standard pneumatic range is 3Psi to 15Psi.

Types of Electric Pressure Transducer :-

- Strain gauge
- Piezoelectric

Strain gauge

- Compressive (length less)
- Tensile (length more)
- Strain = amount of deformation of a body due to applied force
- Use wheatstone bridge or voltage divider

Q. Discuss the ^{working} principle of strain gauges and derive the equation for gauge factor.

Gauge Factor :- (Derivation Class Participation)

Change in Resistance
change in dimension

(Kalsi's Book page 415 - 418)

$$G = \frac{\Delta R / R}{\Delta L / L} = 1 + 2\mu + \frac{\Delta P / P}{\Delta L / L}$$

$$G = \frac{\Delta R / R}{\Delta L / L} = \boxed{1 + 2\mu}$$

Gauge factor for metal is low.

Gauge factor for semiconductor is high.

Inbuilt temperature compensation circuit is required for Germanium & Silicon types of strain gauges.

→ Unbonded strain gauge transducers can be corroded and can be affected by environment causing errors.

Semiconductor strain Gauges

→ Silicon is used more because :

- i) early available
- ii) Thermal stability
- iii) easy doping

For Temperature Compensation :-

Single gauge :- Active Gauge
 Dummy Gauge

For 2, 4, 6 ... gauges not needed.

(ϵ) Strain $\rightarrow$ Input
 v_o $\rightarrow$ Output

$$v_o \propto \epsilon$$

Application of Strain gauges

- Force, Torque, Pressure transducers
- Stress analysis
- Bathroom & Kitchen scales
- Load cells in commercial scales, Tank & vessel weighing
- Measurement of deflection angles of outer surfaces (helicopter blades, wing flaps, elevators etc.)

Q. Explain the construction & working of various electric pressure transducers.

Piezoelectric Transducers

- Apply force to generate voltage
- Universal sensor
- Piezoelectric Accelerometer

Piezoelectric materials

- Natural (Quartz, Topaz)
- Synthetic (Barium Titanate BaTiO_3 , Lead Zirconate PbZrO_3 , PVDF Polyvinylidene Fluoride)

Q. Classify different types of piezoelectric materials

Applications —

- Low Frequency Devices (Force, position, spark generator, acceleration)
- RF Devices (Ultrasonic generator) Receiver / Sonar (Radio Frequency) / High Frequency
- Audio Devices (Mic heads, loud speaker)
- Non Destructive Testing
 - ↳ Ultrasonic testing : To check internal irregularities, cracks in materials.

Differential Pressure Cell (DP Cell)

→ measures differential pressure b/w two inputs.

Q. Measure vacuum pressure

A. Ionisation Gauge
Pirani Gauge

→ Q. Discuss the different techniques of vacuum pressure measurement.

Pirani Gauge

→ heated wire is placed in chamber of gas, thermal conductivity of the gas depends on its pressure.

→ pressure measurement converted to temperature measurement

→ Or, Temperature change causes change in resistance.

Flow Measurement

→ Flow rate = $\frac{\text{volume of substance}}{\text{unit time}}$

→ Turbines, filling a bottle in factory.

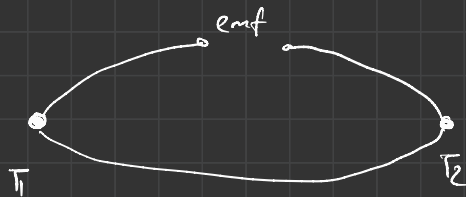
Thermistors vs RTD

- i) Non linear characteristic — Thermistors
Linear characteristics — RTD
- ii) Thermistors have high sensitivity
- iii) Thermistors have low range of measurement.
- iv) Cost and size of thermistors is very less as compared to RTD.
- v) Customized fabrication of thermistors is possible.

Thermo coupling

Thermoelectric effect is

- i) Seebeck effect
- ii) Peltier effect



* Insulation in RTD to prevent corrosion and noise immunity

Q. State the need of temperature compensation in thermocouples and discuss different methods of temperature compensation.

UNIT-3

PROGRAMMABLE LOGIC CONTROLLER

Q. Compare PLC with RLC
or

List advantages of PLC over RLC

Q. Compare PC with PLC
(Computer)

Q. Classify PLC based on structure and types.

Q. Compare continuous processes with discrete processes.

* Watch the PLC experiments from VLabs ...
for M2

engineertech.org → PLC

Q. Architecture of PLC (Long Question)

Q. State the important functions of PLC I/O modules.

UNIT- 4

- Q. Discuss in detail the memory organization of PLC.
- Q. Explain with example how the memory in the PLC is organized.
- Q. How the program is executed in PLC.
- Q. Discuss the steps involved in PLC program execution. State the factors on which the program execution time depends.
or
Explain the steps involved in processor scan cycle.
- Q. Discuss in detail the program scanning process in PLC.
or
Define scan time. State the factors which influence scan time.
- Q. Explain with working analog input and output.
- Q. State the criteria for selecting PLC for the particular application.
- Q. Discuss with example the I/O addressing scheme in PLC.

Programming a PLC

Exp 5

Software :- IndraWorks Engineering

Steps:-

1. File → new project

Name , Compatibility :- 14V22 PS , Folder

2. Add Controller

Library → MLC → L2S → Drag it to project explorer

Next ← Comment ← Name ←

Need matching firmware (otherwise communication error)

Here Firmware version : MLC13VRS

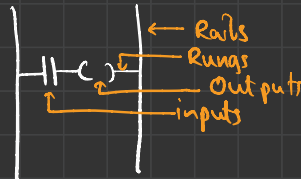
Firmware release : FWA-ML2S*-ML*-13V24.3

Next

Extended Settings

Programming language :- LD (Ladder Diagram)

Ladder Diagram



Execution:- Left → Right Top ↓ Bottom

Graphic
Based

FBD (Function Block Diagram)
Blocks

CFC (Continuous Flow charts)

SFC (Sequential Flow charts)
Chemical industries

Text
based

ST (Structured Text)
Text based

IL (Instruction List)
Text based

function package → next

Interfaces
(Additional components)
↓
Finished

3. Inline → right click → Add → module × 4

RIBIL24DI16

RIBIL24DO16

RIBILA12SF

RIBILA02UBP

Exp 6

AIM :- a) Demonstrate the working of PLC relay logic instructions.

b) To implement the ladder logic program for different types of logic gates & to verify their operations

a) Relay Logic Instruction

b) 1. Right click Indica Motion MLC 1



Simulation

2. To remove door

Project → computer version



3.4.5. 127 (13V16 P9)



Build

3. Expand Logic → Expand Application



Plc Prog



Put logic

Variable

- Name shouldn't be similar to any standard function block (Ex: Ton, Toff, stop...)
- Select proper Type of Variable. (Bool, Time, Int)
- Name in sequential order
- Local Variable, Global Variable

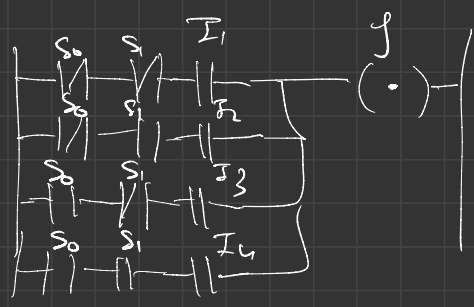
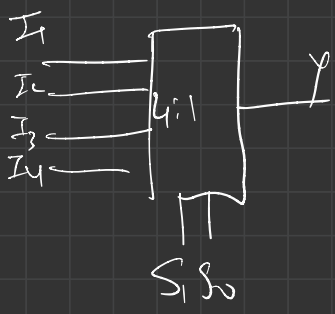
Var Window

Scope :- Var_Globl : Global
 Var : Local
 Var-Input
 Var-Output

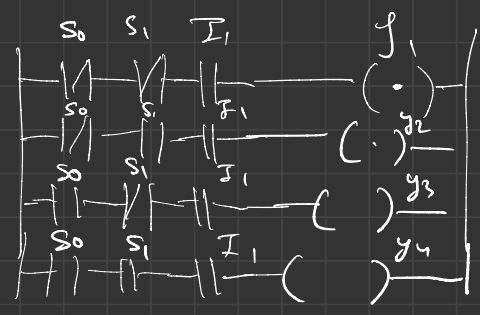
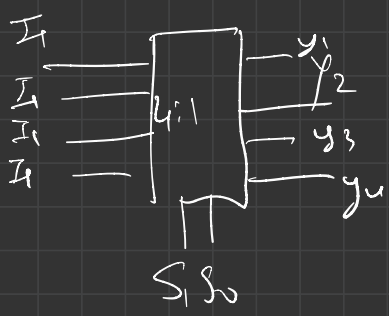
Name → Type (Bool, Int, Byte)

(False) Initialization ← (Plc Prog) Object
→ Address (%IX0.0)

Multiplexer (1:4)

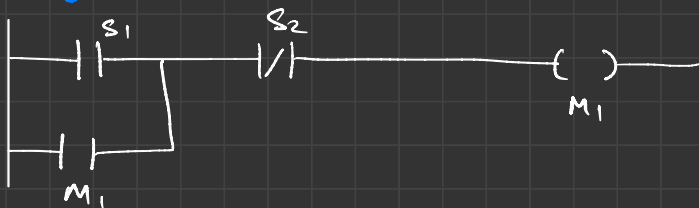


Demultiplexer (4:1)

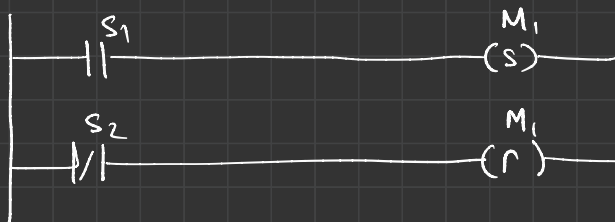


- Q. State various rules to be followed for writing the PLC ladder logic program.
- Q. State the steps to be followed for designing PLC ladder program.
- Q. List different types of PLC ladder logic instruction

Latching (Start-stop mechanism)



Set - Reset Latching

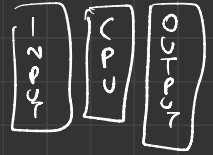


Class Participation

- Q. Industrial Electric Drive → Applications → 5 Applications
(Change program to Bosch Rexroth)

PLC Ladder Logic Question

1. Addressing table (I/O Ports)
2. Program
3. Interfacing diagram
4. Very short description



Q. Timer, Counter, Arithmetic Operations

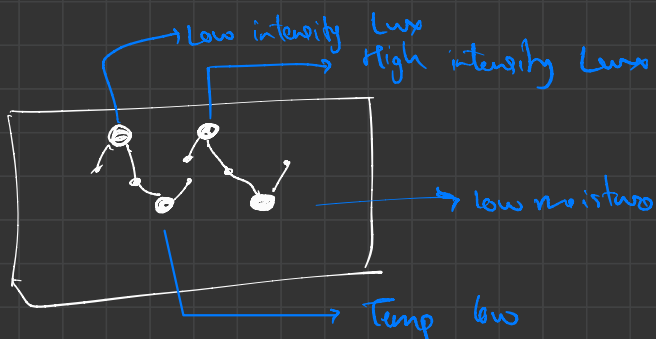
Q. Applications of Ladder Timers in PLC
↳ functions

Q. Discuss with neat timing diagram ON Delay Timer, OFF Delay Timer, Pulse Timer.

Q. Compare PLC timer instructions with counters.

Basket 1

1. Industrial Robotics & Motion Control
modelling, simulation,
speed, motion, torque,
simulations (Pid Control)
2. Microelectromechanical
Systems
Fabrications
3. Fluid Power Automation
Pneumatic systems,
hydraulic actuation systems



TL $\rightarrow$ 11
 BL $\rightarrow$ 10
 TR $\rightarrow$ 9
 BR $\rightarrow$ 6

Dark $\rightarrow$ High

Light $\rightarrow$ Low (650-700)

600

